Transformations of Trigonometric Functions

Exam code: 9MA0

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Updated on 16 February 2026

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Help me understand

Transformations of trigonometric functions

What are transformations of trigonometric functions?

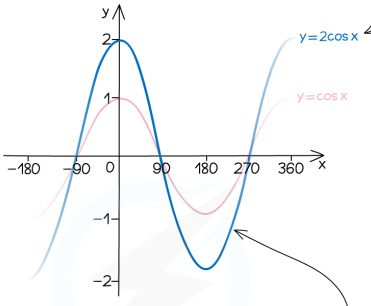
- You should already have a good idea about **translating**, **stretching** and **reflecting** basic functions (see **Transformations of Functions**)
- The basic principles are exactly the same for transforming any trigonometric graph
  - $y = n \cos x$ gives a **vertical stretch or squash**
  - $y = \sin nx$ gives a **horizontal stretch or squash**
  - $y = \tan (x + c)$ gives a **horizontal translation**

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

MAKING IT TWICE AS TALL



STEP 1: SKETCH THE ORIGINAL sin, cos OR tan FUNCTION
STEP 2: PICK KEY CO-ORDINATE POINTS TO PLOT THE NEW GRAPH
STEP 3: SKETCH IN THE NEW TRANSFORMED GRAPH

FOR A VERTICAL STRETCH (OR SQUASH)

- IF $n>1$, THE GRAPH IS **STRETCHED VERTICALLY** BY A SCALE FACTOR OF n
- IF $0<n<1$, THE GRAPH IS **SQUASHED** BY THAT SCALE FACTOR
- IF $n<0$, THE GRAPH IS ALSO **REFLECTED** IN THE x-AXIS
- REMEMBER THE x COORDINATES STAY THE SAME; y COORDINATES ARE MULTIPLIED BY n

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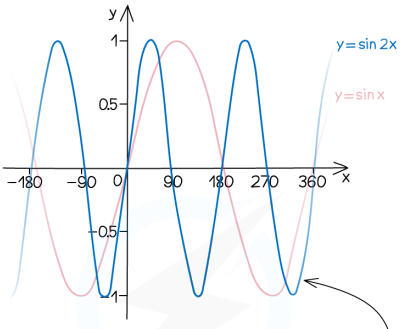
$y=\sin nx$ GIVES A **HORIZONTAL STRETCH** OR **SQUASH**

THE GRAPH OF $y=\sin x$ **SQUASHED HORIZONTALLY** BY A FACTOR OF 2 MEANING IT REPEATS ITSELF TWICE EVERY 360°



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FOR A HORIZONTAL STRETCH (OR SQUASH)

- IF $n>1$, THE GRAPH IS **SQUASHED HORIZONTALLY** BY A SCALE FACTOR OF n
- IF $0<n<1$, THE GRAPH IS **STRETCHED** BY THAT SCALE FACTOR
- IF $n<0$, THE GRAPH IS ALSO **REFLECTED** IN THE y-AXIS
- REMEMBER THE y COORDINATES STAY THE SAME; x COORDINATES ARE MULTIPLIED BY $\frac{1}{n}$

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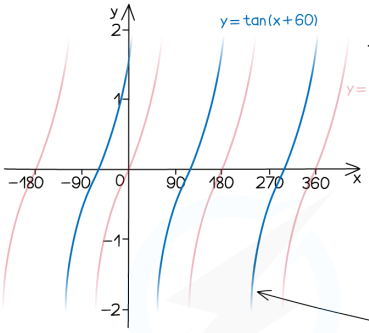

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FOR A HORIZONTAL TRANSLATION

- IF $c>0$, THE GRAPH IS **TRANSLATED LEFT** BY c
  e.g. $\sin(x+40)$ IS $\sin x$ MOVED 40 LEFT
- IF $c<0$, THE GRAPH IS **TRANSLATED RIGHT** BY c
  e.g. $\cos(x-20)$ IS $\cos x$ MOVED 20 RIGHT
- REMEMBER THE y COORDINATES STAY THE SAME; x COORDINATES WILL BE $\pm c$

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Examiner Tips and Tricks

- Always sketch with a pencil and draw a smooth curve.
- When you sketch the transformation of a graph, be sure to indicate the new coordinates of any points that are marked on the original graph.
- Try to indicate the coordinates of points where the new graph intersects the axes.

Worked Example

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